HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Coordinated multidisciplinary environmental inputs for a regional infrastructure delivery programme, integrating acoustic, air quality, and ecology assessments across multiple project phases.
- Supported the preparation of environmental assessment reports by compiling data from various technical specialists and ensuring adherence to reporting standards.
- Collaborated with clients and design teams to identify environmental constraints and develop mitigation solutions for sustainable energy generation and transmission projects.
- Assisted in post-planning consent support, including site surveys and data collection for construction and operational phases of infrastructure developments.
- Developed Python scripts to automate data aggregation from diverse sources, reducing manual compilation effort for environmental impact reports by approximately **30%**.
- Designed and implemented data validation checks using **SQL** across PostgreSQL and Oracle databases, ensuring data integrity for environmental reporting databases.
- Containerised data processing tools using **Docker**, enabling consistent execution environments for environmental data analysis pipelines on AWS EC2 instances.

SKILLS

Environmental Impact Assessment – Environmental Impact Assessment, ESIA, ESDD, Planning, Climate Change, Sustainability, Resilience, Stakeholder Engagement, Acoustics, Air Quality, Ecology, Geospatial & Data Services, Heritage, Landscape, Water Environment.

Data Analysis & Reporting – Power BI, Tableau, R, Python (pandas, NumPy, scikit-learn), SQL (PostgreSQL, MySQL, Oracle), Data Warehousing, ETL, Business Intelligence, DAX, Power Query.

Software Development – Java, Spring Boot, REST APIs, Python scripting, Flask, FastAPI, Docker, Linux, Git, Jenkins, Jira, Confluence.

Cloud Platforms – AWS (EC2, S3, Lambda, CloudWatch).

Project Management & Support – Agile/Scrum, Change Management, Incident Response, SLA Management, Production Support, Regulatory Compliance, GDPR.

PROJECTS

Environmental Data Aggregation and Reporting Platform (HCL Technologies)

- Developed a **Python**-based data pipeline to ingest and aggregate environmental monitoring data from multiple disparate sources, including sensor logs and client-provided datasets.
- Engineered automated reporting functionalities using **Power BI**, creating interactive dashboards that visualised key environmental metrics and trends for operational stakeholders.
- Implemented data validation routines in **SQL** and Python to ensure the accuracy and consistency of the aggregated environmental data, reducing data-related discrepancies by an estimated **20%**.

Infrastructure Project Environmental Constraint Mapping (HCL Technologies)

- Created a geospatial data processing workflow using **Python** and **SQL** to map environmental constraints (e.g., ecological zones, noise sensitivity areas) for infrastructure projects.
- Designed a system to integrate data from various environmental disciplines (e.g., ecology, acoustics) into a unified format for use in impact assessment reports.
- Collaborated with environmental consultants to refine data requirements and ensure the output maps accurately reflected project-specific environmental considerations.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)